

Day 29: End-to-End Professional BI Project

📌 Definition to Remember

Building a production-grade Business Intelligence dashboard requires executing the complete lifecycle: raw data ingestion, dimensional star schema modeling, core DAX metric development, executive UI design, and cloud deployment.

1. Project Scenario: Global Enterprise Executive Sales Dashboard

- **Objective:** Enable the C-Suite and regional directors to monitor \$50M+ in annual revenue, gross margins, customer retention, and regional performance.

2. Step-by-Step Implementation Roadmap

Step 1: Power Query ETL

- Connect to raw multi-table CSV and SQL staging extracts.
- Unpivot monthly targets; clean and standardize product categories.
- Build contiguous **Dim_Date** dimension with Fiscal Year logic.

Step 2: Star Schema Modeling

- Link **Fact_Sales** and **Fact_Targets** to **Dim_Product**, **Dim_Customer**, **Dim_Date**, and **Dim_Region** using strict 1:* single cross-filter relationships.

Step 3: DAX Core Measures Suite

```
// 1. Total Revenue & Margin
Total Revenue = SUM(Fact_Sales[SalesAmount])
Total Cost = SUM(Fact_Sales[CostAmount])
Gross Profit = [Total Revenue] - [Total Cost]
Margin % = DIVIDE([Gross Profit], [Total Revenue], 0)

// 2. Target Variance
Target Revenue = SUM(Fact_Targets[TargetAmount])
Variance to Target $ = [Total Revenue] - [Target Revenue]
Target Achievement % = DIVIDE([Total Revenue], [Target Revenue], 0)

// 3. Time Intelligence
Revenue PY = CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(Dim_Date[Date]))
YoY Growth % = DIVIDE([Total Revenue] - [Revenue PY], [Revenue PY], 0)
```

Step 4: UI/UX Dashboard Layout

- **Header:** Executive summary cards (Revenue, Gross Margin, Target Achievement %, YoY Growth %).

- **Left:** Interactive sidebar filter drawer.
- **Center:** Monthly Revenue vs Target combo chart & Waterfall variance bridge.
- **Bottom:** Top 10 Product Matrix & Regional performance filled map.

Step 5: Service Deployment & RLS

- Configure Dynamic RLS by Regional Manager.
- Publish to Production Workspace, set up daily 7:00 AM Gateway scheduled refresh, and publish the Power BI App.

★ Must-Master Skills & Interview Points

1. Ability to walk an interviewer through the entire architecture from raw data to published app.
2. How you handled messy edge cases during ETL and optimized DAX calculation times.
3. Documenting business requirements and stakeholder KPIs.

⚡ Quick Recall

Raw Data → Power Query ETL → Star Schema → DAX Measure Suite → Executive UI → RLS & App Publish